

Smart Queue System - Project Documentation

Project Description

The Smart Queue System is a web-based application designed to streamline queue management in high-traffic service environments, with an initial focus on medical centers and clinics. The system replaces traditional manual or paper-based queue handling with a digital, real-time solution that enhances both operational efficiency and user experience.

In a medical center context, patients can view available clinics, book tickets remotely, and track their position in the queue without the need for physical presence or long waiting times. Meanwhile, administrators and staff can manage queues dynamically, control patient flow, and monitor real-time activity.

Although the primary implementation targets healthcare facilities, the system is designed to be flexible and adaptable to other domains, including government service centers, banking institutions, and retail branches.

Core Features

1. Real-Time Queue Management

The system provides live queue tracking where updates are instantly reflected across all users without page refresh.

2. Ticket Booking System

Users can book tickets for a specific service or clinic and receive a unique queue number.

3. Waiting Time Estimation

The system estimates the expected waiting time based on queue status, helping users manage their time efficiently.

4. Role-Based Access Control

Patients/Clients: View queues and book tickets

Admins/Staff: Manage queues and control flow

5. Responsive Design

A mobile-first interface ensures accessibility across all devices.

Database Schema Overview

1. User Entity

Name

Email

Password (hashed)

Role (Patient / Admin)

2. Queue Entity

Queue Name
Status (Open / Closed)
Created By (Admin)
Current Active Ticket Number

3. Ticket Entity

Ticket Number
User Reference
Queue Reference
Status (Waiting / In Progress / Completed)
Created At

Relationships

A User can have multiple Tickets
A Queue can have multiple Tickets
Each Ticket belongs to one User and one Queue

Conclusion

The Smart Queue System provides a scalable and efficient solution for managing queues across various industries. By combining real-time updates, structured ticketing, and waiting time estimation, it significantly improves both operational workflows and user experience.